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## **EC-6004-CBGS**

### **B.E. VI Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **VLSI Circuits and Systems**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Implement  $2 \times 1$  MUX using 4 CMOS transistor.  
b) Implement 3-inputs CMOS NAND gates and explain its operations.
2. a) Design a modulo-8 binary counter. Use T flip flops in your realization.  
b) A sequential circuit has four flip flops A, B, C, D and input x. It is described by the following state equations.  
 $A(tH) = A'B'CDx + A'B'Cx + ACDx' + AC'D'x'$   
 $B(tH) = A$   
 $C(tH) = B$   
 $D(tH) = C$   
Design the sequential circuit described by the above state equation using J-K flip flop.
3. a) Define asynchronous sequential machine.  
b) Write and explain finite state machine sequence and prove the results with the help of mealy model.

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4.
  - a) MOS transistors could be worked as switch. Verify this statement with the help of suitable example.
  - b) Explain the verbal description of any MOS equivalent state machines.
5.
  - a) State the difference between mealy and Moore machine.
  - b) Give simple example and draw the state transition diagram for them.
6.
  - a) Explain the races and hazards in asynchronous sequential machine.
  - b) Explain the fundamental mode and pulse mode asynchronous sequential machine.
7. Draw the ASM chart for
  - i) AND Gate
  - ii) SR flip-flop
8. Implement the following Boolean functions using a suitable PAL
  - i)  $w(A, B, C, D) = \sum_m(1, 3, 4, 6, 9, 11, 12, 14)$
  - ii)  $z(A, B, C, D) = \sum_m(2, 3, 8, 9, 12, 13)$

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